

Number Theory and Abstract Algebra

① Is 1729 a Carmichael number?

→ A Carmichael number is a composite number n which satisfies the congruence relation:

$$a^n \equiv a \pmod{n}$$

for all integers a that are relatively prime to n .

To prove that, 1729 is a Carmichael number, we need to show that it satisfies the above

condition.

Step 01:

As given, $n = 1729 = 7 \times 13 \times 19$

Let, $P_1 = 7$, $P_2 = 13$ and $P_3 = 19$

Then $P_1 - 1 = 6$, $P_2 - 1 = 12$ and $P_3 - 1 = 18$

Also, $n - 1 = 1729 - 1 = 1728$ which is divisible by

$$P_1 - 1 = 6$$

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Therefore, $n-1$ is divisible by p_1-1 .

Step 02:

Similarly, we can show that $n-1$ is also divisible by p_2-1 and p_3-1 .

Therefore, from the definition of Carmichael numbers and the above discussion, we can conclude that 1729 is indeed a Carmichael number.

② Primitive Root (Generator) of $\mathbb{Z}_{23}$?

Definition: A primitive root modulo a prime

p is an integer a in $\mathbb{Z}_p$ such that every nonzero element of $\mathbb{Z}_p$ is a power of a .

We want to find a primitive root modulo

23, an element $g \in \mathbb{Z}_{23}$ such that the powers

of g generate all non-zero elements of $\mathbb{Z}_{23}$.

let,

$\mathbb{Z}_{23}$ = the set of integers from 1 to 22.

under multiplication modulo 23. since 23 is a prime number,

$$|\mathbb{Z}_{23}^*| = \phi(23) = 22$$

so, a primitive root g is an integer such that:

$$g^k \not\equiv 1 \pmod{23} \text{ for all } k < 22,$$

$$\text{and } g^{22} \equiv 1 \pmod{23}.$$

we check for $g=5$:

- Prime factors of $22 = 2, 11$

- $5^{22/2} = 5^{11} \pmod{23} = 22 \neq 1$

- $5^{22/11} = 5^2 = 25 \pmod{23} = 2 \neq 1$

So, 5 is a primitive root modulo 23.

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③ Is $\langle \mathbb{Z}_{11}, +, * \rangle$ a Ring?

Yes, $\mathbb{Z}_{11} = \{0, 1, 2, \dots, 10\}$ with addition and multiplication modulo 11 is a Ring because:

- $(\mathbb{Z}_{11}, +)$ is an abelian group.

- Multiplication is associative and distributes over addition.

- It has a multiplicative identity.

Since 11 is prime, $\mathbb{Z}_{11}$ is also a field.

So $(\mathbb{Z}_{11}, +, *)$ is a Ring.

④ Is $\langle \mathbb{Z}_{37}, + \rangle, \langle \mathbb{Z}_{35}, * \rangle$ are abelian group?

Answer:

$(\mathbb{Z}_{37}, +)$:

This is an abelian group under addition mod 37. Always true for $\mathbb{Z}_n$ with addition.

$(\mathbb{Z}_{35}, *)$:

This is not an abelian group.

Only the units in $\mathbb{Z}_{35}$ form a group under multiplication. But full $\mathbb{Z}_{35}$ under multiplication includes 0, non-invertibles.

So it's not a group.

⑤ Let's take $p=2$ and $n=3$ that makes the $GF(p^n) = GF(2^3)$ then solve this with polynomial arithmetic approach.

Answer:

Given, $p=2, n=3$

We want to construct the finite field $GF(2^3)$ which has $2^3=8$ elements.

Step 1: choose an irreducible polynomial.

To build $GF(2^3)$ select an irreducible polynomial of degree 3 over $GF(2)$.

A common choice is:

$$f(x) = x^3 + x + 1$$

Step 2: Define the field elements, Every element of $GF(2^3)$ can be express as a polynomial with degree less than 3 and coefficients

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in $GF(2)$:

$$\{0, 1, x, x+1, x^v, x^v+1, x^v+x, x^v+x+1\}$$

Step 3: Define addition and multiplication.

- Addition is performed by adding corresponding coefficients modulo 2,

$$x^v+x=0, x^v+1=x^v+1$$

- Multiplication is polynomial multiplication followed by reduction modulo,

$$f(x) = x^3+x+1$$

$$x^3 \equiv x+1 \pmod{f(x)}$$

Example calculations:

- $x \cdot x = x^v$ (no reduction needed)
- $x \cdot x^v = x^3 = x+1$
- $(x+1) \cdot x = x^v+x$

Thus, $GF(2^3)$ is a field with 8 elements and well defined addition and multiplication.